FEBRIS CO2 PAYLOAD description

This chapter describes the structure of the telemetry data. The number of bytes in the payload depends on the configuration of the sensor. The structure of the data depends on which sensors are present in your product. In principle, each version has a header. This has information about the version and status of the sensor and also contains a master measured value. This measured value can simply be assumed to be the current measured value. The header is followed by an additional payload part that contains further information, such as special parameters for the measuring principle or settings for position and opening detection. Not every sensor has an additional payload part.

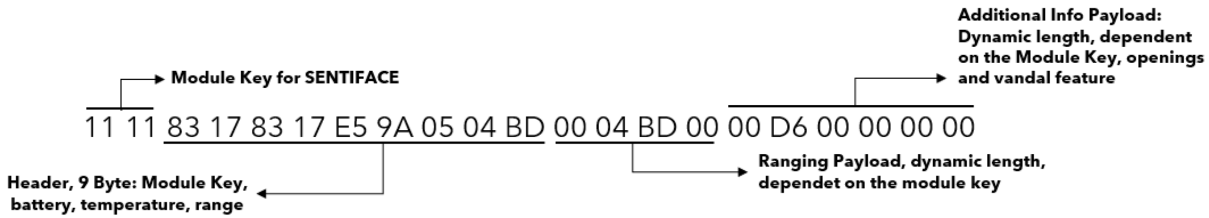
Terms	Description
Byte No.	Byte number starting at 1
Alias	Comprehensible name of the variable
Description	Description of the variables
Designation	Designation in the Dataconverter
Unit	Unit of the variables
Datahub .json Schlüssel	Key in the .json when using the data hub

STRUCTURE OF THE MODULE KEY

Byte 1		Byte 2	
Bit 7 – 4	Bit 3 – 0	Bit 7 – 4	Bit 3 – 0
Basis-ID-Module e.g. Sentiface, Senticom, Sentivisor	Major Version (SW/HW version)	Minor version (SW/HW version)	Product version (Sensors, e.g. TH, THL, ACC, …)

The module key of the SENTIFACE module can be taken from the first 2 bytes of each uplink. The module key is required for the downlink.

UPLINK EXAMPLE (NOT APPLICABLE FOR FEBRIS CO2)



The following payload example is given for the Helios pressure sensor:

`11 11 FE 1A D5 95 06 03 00 23 BE`

Bytes	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**
HEX	11	11	FE	1A	D5	95	06	03	00	23	BE
Description	Module key	Module key	Uplink Counter	Battery voltage	Battery voltage	Tempera-ture	Alarm Flag	Alarm Flag	Measurement status	Pressure mbar	Pressure mbar

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FEBRIS CO2 PAYLOAD DESCRIPTION

Byte number	Alias name	Description	Unit	Designation in the payload decoder	Module key	Datahub .json key
1 - 2	Module key	Byte 1 • 4-bit MSB: Module code. For Mioty only telemetry module "Sentiface", i.e. always 0x1. • 4 Bit LSB: Major version. FW with the same major version but a larger minor version remain compatible with older decoders. However, the number of bytes may change! Byte 2 • 4-bit MSB: minor version. New SW versions can gradually add new telemetry to the data package. • 4 bit LSB: Sub/product version. In Helios, these bits encode the specific configuration of the connected sensors.		module_key	XXX1	module_key
3	Uplink counter	Protocol independent uplink counter that reaches a maximum value of 255. After this, the counter starts again at 0.		uplink_counter	XXX1	uplink_counter
4 - 5	Battery voltage	Current battery voltage in millivolts //Voltage in millivolts battery_voltage: (dataBytes[3] << 8 | dataBytes[4])/1000.0	mV	battery_voltage	XXX1	battery_voltage
6 - 7	Interne Temperaur	Temperature of the sensor: Internal_temperature = ((bytes[5] << 8) | bytes[6]) / 10 - 100	°C	internal_ temperature	XXX1	internal_ temperature

Byte number	Alias name	Description	Unit	Designation in the payload decoder	Module key	Datahub .json key
8	Relative humidity	Relative humidity in %RH	%RH	humidity	XXX1	humidity
9 - 10	Dew point	Dew point in °C	°C	dew_point	XXX1	dew_poin
11 - 12	Air pressure	Air pressure in hPa	hPa	pressure	XXX1	pressure
13 - 14	CO2	CO2 concentration in ppm	ppm	co2_ppm	XXX1	co2_ppm
15	Alarm	Status of the gas sensor: 0: 0 - 1000ppm (green) 1: 1000 - 2000ppm (yellow) 2: 2000 - 5000ppm (red) Analog to the traffic light, adjustable limits	°C	alarm	XXX4	alarm

IAQ INDEX

The IAQ index (Indoor Air Quality Index) is an indicator that assesses the quality of indoor air and is often used to describe the health status of indoor air. It is typically calculated on the basis of measurements of various air pollutants.

OBJECTIVE OF THE IAQ-INDEX

- **- Health protection**: The IAQ index indicates how safe or healthy the air is for the people in the room.
- **- Orientation aid**: It enables a simple and comprehensible interpretation of the air quality, often in the form of a scale value or a color classification.

IMPORTANT PARAMETERS FOR THE IAQ-INDEX

The IAQ index is based on various factors and measurements, including:

1. **Carbon dioxide (CO_2):**

    - High concentrations can lead to fatigue, concentration problems and discomfort.
    - Typical limit value: 400-1000 ppm (parts per million) is considered acceptable.

2. **Volatile organic compounds (VOC):**

    - Chemical substances that are released from paints, cleaning agents, furniture or electronic devices.
    - At high concentrations, VOCs can cause headaches, irritation or long-term health problems.

3. **Particulate matter (PM2.5/PM10)**:

    - Fine particles produced by smoke, cooking fumes or dust.
    - Particulate matter can cause respiratory diseases and other health problems.

4. **Carbon monoxide (CO):**
    - A toxic gas produced by incomplete combustion.
    - Even low concentrations can be dangerous.

5. **Relative humidity and temperature:**
    - These values influence the well-being and growth of mold fungi or bacteria.

CLASSIFICATION OF THE IAQ-INDEX

Many IAQ systems use a numerical or color scale to represent air quality. For example:

IAQ value	Air quality	Description
0-50	Excellent	No health risks.
51-100	Good	Low risks for sensitive persons.
101-150	Moderate	May be problematic for sensitive individuals.
151-200	Unhealthy	Possible adverse effects for the general population.
201-300	Very unhealthy	Health problems with prolonged exposure.
301+	Dangerous	Serious health risks.

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EXAMPLES DECODING

Payload: 11 11 CE 14 8E 07 EE 1D 00 02 03 D6 01 D1 00

Bytes	1	2	3	4	5	6	7	8	9	10	11
HEX	11	11	CE	14	8E	07	EE	1D	00	02	03
Description	Module key	Module key	Uplink Counter	Battery Voltage	Battery Voltage	Internal temperature	Internal temperature	Rel. humidity	Dew point	Dew point	Air pressure

Bytes	12	13	14	15
HEX	D6	01	D1	00
Beschreibung	Air pressure	CO2	CO2	Alarm


```
"uplink_message": {
  "session_key_id": "AYY660iFsk5LInv+xd6AVg==",
  "f_port": 1,
  "f_cnt": 211294,
  "frm_payload": "ESPOFI4Esx0D1gHRAAABZg==",
  "decoded_payload": {
    "alarm": 0,
    "base_id": 1,
    "battery_voltage": 5.262,
    "co2_ppm": 465,
    "dew_point": 2,
    "humidity": 29,
    "internal_temperature": 20.299999999999997,
    "major_version": 1,
    "minor_version": 2,
    "networkBaseType": "lorawan",
    "networkSubType": "tti",
    "pressure": 982,
    "product_version": 3,
    "up_cnt": 206
  },
```